USFQ POSGRADOS

# cGAN for MFTR Wireless Fading Distribution

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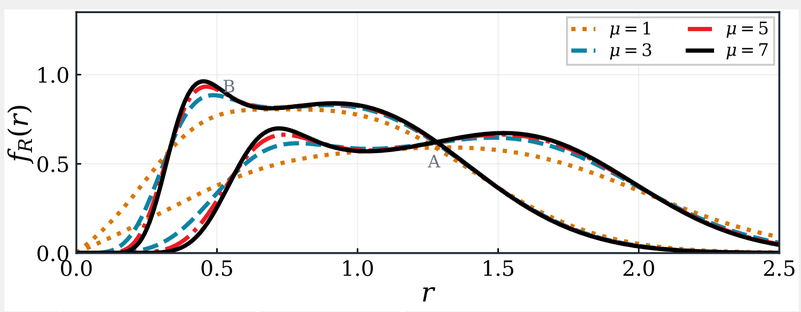
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## Introduction

The **MFTR** model [1] unifies FTR and $\kappa$-$\mu$ shadowed fading via $\{m, \mu, K, \Delta, \omega\}$. Its PDF transitions from **unimodal to bimodal** as $\mu$ grows.

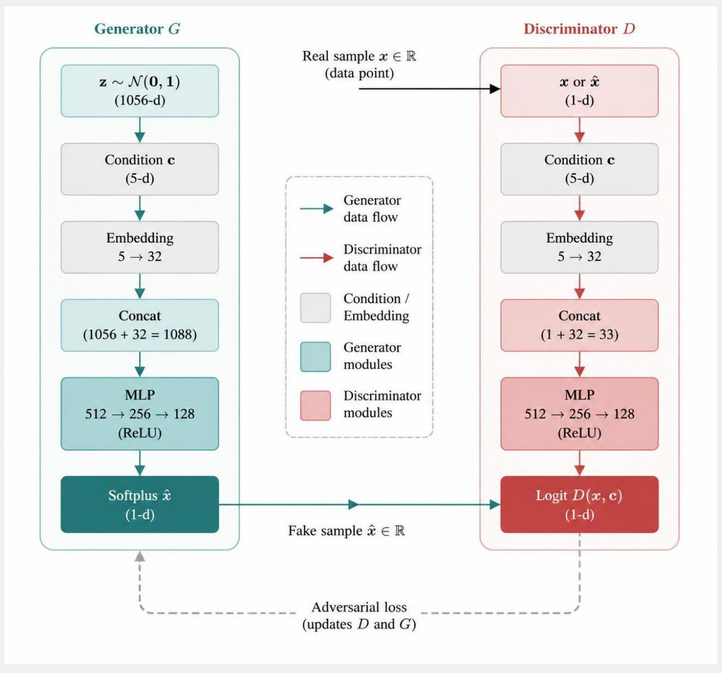
**Goal:** train a **cGAN** on synthetic MFTR amplitudes conditioned on all parameters and evaluate interpolation/extrapolation.

## MFTR Distribution




$f_R(r)$ for $\mu \in \{1, 3, 5, 7\}$, two parameter sets A and B. Bimodality grows with $\mu$.

## Architecture

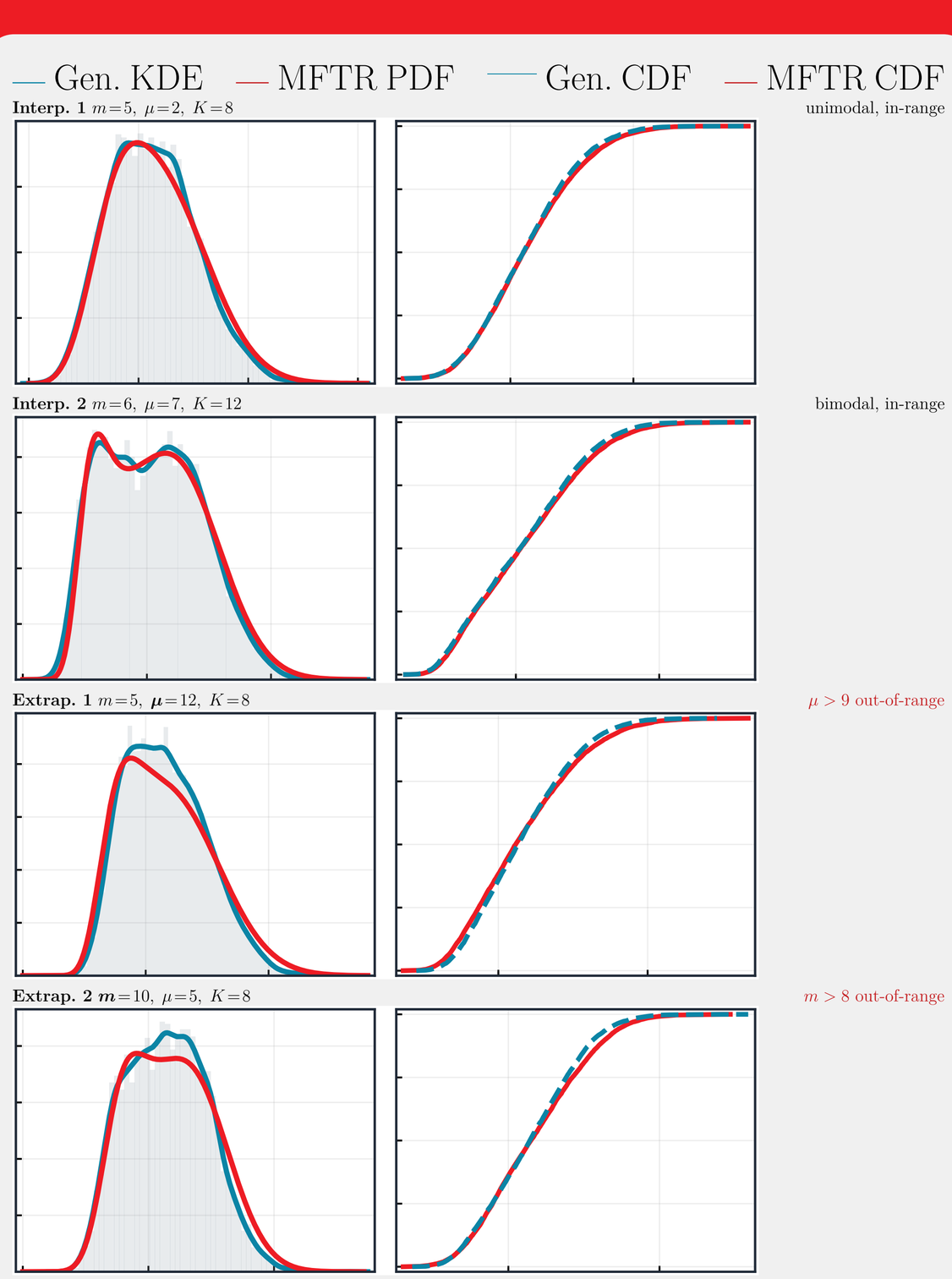



Shared embedding: Linear–ReLU–Linear ($5 \to 32$), $\mathbf{c}$ normalized to $[-1, 1]$. Adam $\beta_1 = 0.5$, $\mathrm{lr}_G = 2 \times 10^{-4}$, $\mathrm{lr}_D = 4 \times 10^{-4}$, batch 1024, update $G$ every 3 discriminator steps.

## Training Pipeline

1) Sample MFTR parameters $\{m, \mu, K, \Delta, \omega\}$
2) Generate amplitudes $x \sim \mathrm{MFTR}$
3) Build conditional dataset $(x, \mathbf{c})$
4) Adversarial training with cGAN
5) Evaluate interpolation/extrapolation and export plots

## Results: Interp. vs. Extrap.

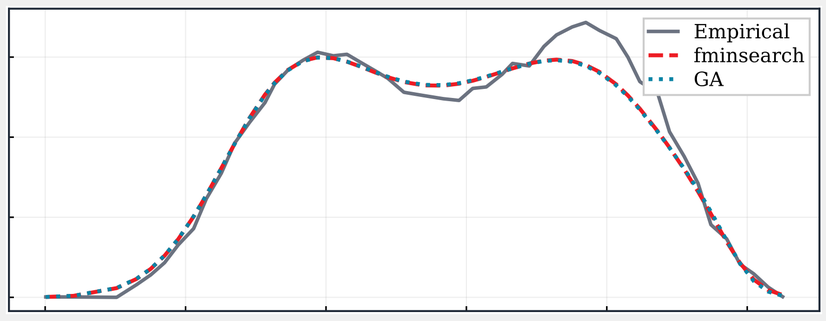



Each panel: **left** density (hist. + KDE + MFTR PDF), **right** empirical CDF. Interp. matches theory closely across unimodal and bimodal shapes. Extrap. degrades gracefully: similar median but larger worst-case errors (Wasserstein $\approx$0.11 vs. $\approx$0.05).

## Next Steps

Prioritized next step: transfer the full pipeline to **real wireless measurements**.

Six empirical scenarios with MFTR fitted parameters are ready. This poster prioritizes model behavior and synthetic-to-theory agreement.




Scenario 1: empirical (—), fminsearch (–), GA ($\cdots$).